Longbiao CHEN

Ph.D. | Researcher
Crowd Intelligence and Spatial Computing
Ubiquitous and Embodied Intelligence

Email: longbiaochen@acm.org
Homepage: longbiao.crowdsensing.cn

Education

2015–2018 **Ph.D. Degree, Sorbonne University, France**
Supervisors: Prof. Daqing ZHANG (Fellow, IEEE) and Prof. Thi-Mai-Trang NGUYEN

2010–2016 **Ph.D. Degree, Zhejiang University, China**
Supervisor: Prof. Gang PAN

2006–2010 **B.E. Degree, Zhejiang University, China**
Chu Kochen Honors College, Computer Science

Academic Appointments

1. 2021.08 Promoted to Associate Professor, Department of Computer Science and Technology, School of Informatics, Xiamen University (former appointment).
2. 2018.08-2021.07 Assistant Professor, Department of Computer Science, School of Informatics, Xiamen University, China.
3. 2016.10-2018.07 Assistant Researcher, Networks and Multimedia Department, Institut Mines-Telecom, France.

Research Interests

Research in crowd intelligence, embodied intelligence, spatial computing, ubiquitous computing and domain-specific large language models, with applications in urban governance, public safety, robotic services and healthcare. Ph.D. degrees from Zhejiang University and Sorbonne University.

Research Experience

1. **Public Transportation System Planning and Management**
   Planning and managing public transportation systems of a large city is challenging for urban authorities. In this work, we take bike sharing system as an example to explore the potentials of leveraging urban big data to address public transportation challenges. In the bike station planning phase, we identify three important factors related to the bike usage demand of an area, i.e., the POI distributions, the human flow intensities, and the demographic characteristics. We then fuse the corresponding urban big data to model the bike usage demand of each area, and recommend optimal area for placing bike stations. In the bike station management phase, we model the bike usage pattern of each area leveraging relevant urban big data, such as weather conditions and urban events, and predict the station outage (e.g., being empty) to help designing optimal bike balancing strategy between stations. (This work has received **the best paper candidate award of ACM UbiComp'15** and has been accepted by **ACM UbiComp'16**)

2. **Container Port Performance Measurement and Comparison**
   Container ports are the logistics centers of international trade. Being able to measure the port performance is key to port operation and logistics optimization. Traditional port performance data collected from port authorities and shipping companies are usually delayed and vary in granularities, hindering the measurement and comparison of international ports. In this work, we propose a framework to detect ship arrivals in container ports and estimate the number of containers handled by leveraging ship GPS trajectories and port facility parameters. We further derive a set of performance metrics (e.g., terminal productivity and monthly port throughput) to measure and compare the top-100 container ports worldwide. (This work has been published in **ACM UbiComp'14** and **IEEE T-ITS**)

3. **Urban Events Detection and Characterization**
   Urban events (such as festivals and concerts) usually attract large crowds of people, leading to potential traffic congestion, noise pollution, and security risks. In order to better manage these events, urban authorities need to understand the human flow directions, crowd activities, and even crowd sentiment during events. In this work, we fuse human mobility traces (e.g., taxi GPS trajectories) and crowd activity data (e.g., location based social network user check-ins) to automatically detect urban events from the significant and unusual human flow they cause, and characterize the human mobility trends and the crowd activities involved in these events. (This work has been published in **IEEE T-HMS**)

4. **Twitting Coffee Machine: A Fusion of Social Network and Internet of Things**
   With the advances of smart home technologies, an increasing number of home appliances are connected to the Internet. In this work, we take a step forward to connect home appliances to the social network of human. In this way, the working status of these appliances can be monitored through their tweets, and the appliances can be controlled by natural commands. We successfully connect a coffee machine to Sina Weibo, the Chinese Twitter. It gains 30,000 followers within 48 hours, and receives broad media coverage including China Daily. (This work has been published in **IEEE UIC'12**)

Publications

Selected Publications

1. Guofeng Luo, Junxiang Ji, Yongyi Wu, Ruixiang Luo, Jiaru Wang, Lijuan Weng, Zhuohan Ye, Chenhui Lu, Dingqi Yang, Cheng Wang, **Longbiao Chen**. UVX-ray: Urban Village Safety Risk Diagnosis Leveraging Multi-Source Urban Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025. **[CCF-A]** [Online]

2. Yayao Hong, Liyue Chen, Leye Wang, Xiuhuai Xie, Guofeng Luo, Cheng Wang, **Longbiao Chen**. STKOpt: Automated Spatio-Temporal Knowledge Optimization for Traffic Prediction. *Proceedings of the ACM Web Conference 2025*, 2025. **[CCF-A]** [Online]

3. Guofeng Luo, Lijuan Weng, Yunqian Li, Yilu Sun, Yayao Hong, Yongyi Wu, Ruixiang Luo, Leye Wang, Cheng Wang, **Longbiao Chen**. FireExpert: Fire Event Identification and Assessment Leveraging Cross-Domain Knowledge and Large Language Model. *IEEE Transactions on Mobile Computing*, 2025. **[CCF-A / JCR-I]** [Online]

4. Yufei Wang, Wenting Zeng, Changzhen Liu, Zhuohan Ye, Jiawei Sun, Junxiang Ji, Zhihan Jiang, Xianyi Yan, Yongyi Wu, Yigao Wang, Dingqi Yang, Leye Wang, Daqing Zhang, Cheng Wang, **Longbiao Chen**.

CrowdBot: An Open-Environment Robot Management System for On-Campus Services. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2024. **[CCF-A]** [Online]

5. Yayao Hong, Hang Zhu, Tieqi Shou, Zeyu Wang, Liyue Chen, Leye Wang, Cheng Wang, **Longbiao Chen**. STORM: A Spatio-Temporal Context-Aware Model for Predicting Event-Triggered Abnormal Crowd Traffic. *IEEE Transactions on Intelligent Transportation Systems*, 2024. **[CCF-B / JCR-I]** [Online]

6. **Longbiao Chen**, Xin He, Xiantao Zhao, Han Li, Yunyi Huang, Binbin Zhou, Wei Chen, Yongchuan Li, Chenglu Wen, Cheng Wang. GoComfort: Comfortable Navigation for Autonomous Vehicles Leveraging High-Precision Road Damage Crowdsensing. *IEEE Transactions on Mobile Computing*, 2023. **[CCF-A / JCR-I]** [Online]

7. Tieqi Shou, Zhuohan Ye, Yayao Hong, Zhiyuan Wang, Hang Zhu, Zhihan Jiang, Dingqi Yang, Binbin Zhou, Cheng Wang, **Longbiao Chen**. CrowdQ: Predicting the Queue State of Hospital Emergency Department Using Crowdsensing Mobility Data-Driven Models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2023. **[CCF-A]** [Online]

8. Zhihan Jiang, Xin He, Chenhui Lu, Binbin Zhou, Xiaoliang Fan, Cheng Wang, Xiaojuan Ma, Edith C.H. Ngai, **Longbiao Chen**. Understanding Drivers' Visual and Comprehension Loads in Traffic Violation Hotspots Leveraging Crowd-Based Driving Simulation. *IEEE Transactions on Intelligent Transportation Systems*, 2022. **[CCF-B / JCR-I]** [Online]

9. Hang Zhu, Tieqi Shou, Ruiying Guo, Zhihan Jiang, Zeyu Wang, Zhiyuan Wang, Zhiyong Yu, Weijie Zhang, Cheng Wang, **Longbiao Chen**. RedPacketBike: A graph-based demand modeling and crowd-driven station rebalancing framework for bike sharing systems. *IEEE Transactions on Mobile Computing*, 2022. **[CCF-A / JCR-I]** [Online]

10. **Longbiao Chen**, Chenhui Lu, Fangxu Yuan, Zhihan Jiang, Leye Wang, Daqing Zhang, Ruixiang Luo, Xiaoliang Fan, Cheng Wang. UVLens: urban village boundary identification and population estimation leveraging open government data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2021. **[CCF-A]** [Online]

11. Zhihan Jiang, Hang Zhu, Binbin Zhou, Chenhui Lu, Mingfei Sun, Xiaojuan Ma, Xiaoliang Fan, Cheng Wang, **Longbiao Chen**. CrowdPatrol: A mobile crowdsensing framework for traffic violation hotspot patrolling. *IEEE Transactions on Mobile Computing*, 2021. **[CCF-A / JCR-I]** [Online]

12. **Longbiao Chen**, Thi-Mai-Trang Nguyen, Dingqi Yang, Michele Nogueira, Cheng Wang, Daqing Zhang. Data-driven C-RAN optimization exploiting traffic and mobility dynamics of mobile users. *IEEE Transactions on Mobile Computing*, 2020. **[CCF-A / JCR-I]** [Online]

13. **Longbiao Chen**, Xiaoliang Fan, Leye Wang, Daqing Zhang, Zhiyong Yu, Jonathan Li, Thi-Mai-Trang Nguyen, Gang Pan, Cheng Wang. RADAR: road obstacle identification for disaster response leveraging cross-domain urban data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2018. **[CCF-A]** [Online]

14. **Longbiao Chen**, Jérémie Jakubowicz, Dingqi Yang, Daqing Zhang, Gang Pan. Fine-grained urban event detection and characterization based on tensor cofactorization. *IEEE Transactions on Human-Machine Systems*, 2017. **[CCF-B / JCR-II]** [Online]

15. **Longbiao Chen**, Daqing Zhang, Xiaojuan Ma, Leye Wang, Shijian Li, Zhaohui Wu, Gang Pan. Container port performance measurement and comparison leveraging ship GPS traces and maritime open data. *IEEE Transactions on Intelligent Transportation Systems*, 2016. **[CCF-B / JCR-I]** [Online]
16. **Longbiao Chen**, Daqing Zhang, Leye Wang, Dingqi Yang, Xiaojuan Ma, Shijian Li, Zhaohui Wu, Gang Pan, Thi-Mai-Trang Nguyen, Jérémie Jakubowicz. Dynamic cluster-based over-demand prediction in bike sharing systems. *2016 ACM international joint conference on pervasive and ubiquitous computing*, 2016. **[CCF-A]** [Online]
17. **Longbiao Chen**, Daqing Zhang, Gang Pan, Xiaojuan Ma, Dingqi Yang, Kostadin Kushlev, Wangsheng Zhang, Shijian Li. Bike sharing station placement leveraging heterogeneous urban open data. *2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2015. **[CCF-A]** [Online]
18. **Longbiao Chen**, Daqing Zhang, Gang Pan, Leye Wang, Xiaojuan Ma, Chao Chen, Shijian Li. Container throughput estimation leveraging ship GPS traces and open data. *2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2014. **[CCF-A]** [Online]

Full publication list: `https://longbiao.crowdsensing.cn/publications/`

Patents

1. Deep-learning-based venue flow control method, system and readable medium. 2022112269437
2. Robot task management system based on deep reinforcement learning. 202411116566.0
3. Deep-learning-based urban-village risk assessment and risk-factor localization. 202210493040.9
4. Deep-learning-based urban-village identification and population estimation. 202010873246.5

Selected Awards

2022	**ACM SIGSPATIAL China Rising Star Award**
2015, 2016	**ACM UbiComp Honorable Mention Awards**
2018	First Prize, Fujian Provincial Science and Technology Progress Award
	Multi-source urban transportation sensing and intelligent computing (ranked ninth)
	Huawei Cloud Pioneer Teacher Award

Research Projects

2022–2025	NSFC General Program, Principal Investigator.
2019–2021	NSFC Young Scientists Fund, Principal Investigator.
2019–2021	National defense basic research project, Principal Investigator.
2020–2021	National high-resolution Earth observation project, Principal Investigator.
2018–2021	Natural Science Foundation of Fujian Province, General Program, Principal Investigator.
2020–2021	Xiamen industry-university-research collaborative innovation project, Principal Investigator.

Teaching Experiences

2020–2025 Course Instructor, Software Engineering, Xiamen University
2019–2020 Software Architecture and Application Development, Xiamen University
2018–2024 Pattern Recognition, Xiamen University
2018–2019 Human Computer Interaction, Xiamen University Malaysia

Talks

2015 **ACM UbiComp 2015 (Host: Youngki Lee), Osaka, Japan, September 2015**
Title: *Bike-Sharing Station Placement Leveraging Heterogeneous Urban Open Data*
2015 **IEEE UIC 2015 (Host: Zhiyong Yu), Beijing, China, August, 2015**
Title: Sensing the Pluse of Urban Activity Centers
2015 **SystemX Réunion (Host: Lionel Scremin), Paris, France, July 2015**
Title: *Human Flow Inference on Bike-Sharing Systems*
2014 **ACM UbiComp 2014 (Host: Nigel Davies), Seattle, USA, September 2014**
Title: *Container Throughput Estimation Leveraging Ship GPS Traces and Open Data*

Services

Editor-in-Chief, Big Data Research; editorial board service for Vehicles, Frontiers in Sustainable Cities and JILSA.
Executive committee member, ACM SIGSPATIAL China Chapter; executive committee member, CCF Ubiquitous Computing Committee.
CCF YOCSEF Xiamen: Chair (2021), Vice Chair (2020), Academic Secretary (2019).
Social Activity Chair, ACM UbiComp 2021; Industry Chair, AIoTSys 2025.
Member, BRICS Young Scientist Forum and moderator of its AI session (2022); program committee service for AAAI and IJCAI.
Co-editor: Spatial Data and Intelligence, SpatialDI 2025 proceedings, LNCS 15838, Springer.

Language Skills

English: Fluent **French**: Working knowledge **Chinese**: Native speaker